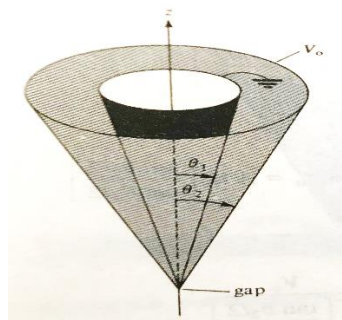
Course PH410 Course coordinator: Alika Khare

Problem Set _03 Due Date: Jan 29, 2025

Note: Please submit the solutions of at least two problems, on wednesday, Jan 29, 2025, during the class. The solutions can be submitted in a group of maximum of four students. Name and Roll no. of all the group members should be written clearly on the top. If solutions are found to be reasonably correct then a total of 10 marks (five marks each) will be awarded and in case of non-submission/wrong solution zero marks. Where ever applicable draw the well labeled diagram along with axis properly marked.

1. A uniformly charged metal sphere carrying a total charge Q is placed in an otherwise uniform electric field $\overrightarrow{E_o}$. Find the potential and electric field outside.

2. Two conducting cones at $\theta_1 = \pi/10$ and $\theta_2 = \pi/6$ having an infinitesimally small hole at r=0 as shown below in the figure. The inner cone is maintained at 0 potential whereas the outer one at 50 V. Find V and **E** in between the cones.



3. Suppose that the surface of a sphere of radius R carries a potential $V_o(\theta)$. Find the potential inside and out side the sphere and hence the expression for surface charge density.
4. In the above problem, Problem 2, if the potential at the surface $V(\theta) = K\sin^2\theta$, ($K$ being a constant) than calculate the potential inside as well as outside the sphere.
5. A spherical shell of radius R carries uniform surface charge density σ_o on the upper hemisphere and that of $-\sigma_o$ on the bottom hemisphere. Find the potential inside and outside the sphere up to a minimum of three non zero terms in both the cases.

6. A conducting sphere of radius a, maintained at a potential V_o, is surrounded by a thin concentric spherical shell of radius b, carrying a surface charge density $\sigma(\theta) = k\cos\theta$, where k is a constant.
   (a) Find the potential in region: (i) $r > b$, and (ii) $a < r < b$.
   (b) Find the induced surface charge density, $\sigma_{ind}(\theta)$, on the conductor.
   (c) What is the total charge of this system? Check that your answer is consistent with the behavior of V at large r.

PH410: ELECTRODYNAMICS-I
Tutorial 3: Potentials

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1. A uniformly charged metal sphere carrying a total charge Q is placed in an otherwise uniform electric field $\vec{E_0}$. Find the potential and electric field outside.

Solution:
The solution for Laplace equation in spherical coordinates, in case of azimuthal symmetry is gven by:

$$V(r,\theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos\theta)$$

Boundary conditions are:

(i) $V = 0$, when $r = R$.

(ii) $V \to -E_0 r \cos\theta$ for $r \gg R$.

Let V_1 be potential inside the sphere and V_2 be potential outside the sphere. At $r = R$, V is continuous, then

$$V_2(r = R) = V_1(r = R)$$

$$\sum_{l=0}^{\infty} \left(A_l R^l + \frac{B_l}{R^{l+1}} \right) P_l(\cos\theta) = 0$$

$$A_l R^l + \frac{B_l}{R^{l+1}} = 0 \Longrightarrow B_l = -A_l R^{2l+1}$$

At a very large distance $r \gg R$, the second term of V_2 inside the parenthesis tends to zero. Therefore,

$$\sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta) = -E_0 r \cos\theta$$

Comparing the coefficient, we could deduce that only one term $l = 1$ is present.

$$A_1 = -E_0$$

$$\begin{aligned} \therefore V_2(r,\theta) &= \left(A_1 r + \frac{B_1}{r^2} \right) \cos\theta = \left(A_1 r + \frac{(-A_1 R^3)}{r^2} \right) \cos\theta \\ &= \left(-E_0 r + \frac{E_0 R^3}{r^2} \right) \cos\theta = -E_0 \left(r - \frac{R^3}{r^2} \right) \cos\theta \end{aligned}$$

Since the metal is charged, a coulomb potential should be considered too. Then the total potential is given by:

$$V(r,\theta) = -E_0 \left(r - \frac{R^3}{r^2} \right) \cos\theta + \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r}$$

2. Two conducting cones at $\theta_1 = \pi/10$ and $\theta_2 = \pi/6$ having an infinitesimally small hole at

$r = 0$ as shown below in the figure. The inner cone is maintained at 0 potential whereas the outer one at $50V$. Find V and E in between the cones.

Solution:
In this configuration the potetntial is depends on θ only, so we may write Laplace equation in spherical coordinates as:

$$\nabla^2 V = \frac{1}{r^2 sin\theta}\frac{d}{d\theta}\left[\sin\theta\frac{dV}{d\theta}\right] = 0$$

$$\frac{d}{d\theta}\left[\sin\theta\frac{dV}{d\theta}\right] = 0 \Longrightarrow \sin\theta\frac{dV}{d\theta} = A \Longrightarrow \frac{dV}{d\theta} = \frac{A}{\sin\theta}$$

Integrating, we get

$$V = \int \frac{d\theta}{\sin\theta} = A\int \frac{d\theta}{2\cos(\theta/2)\sin(\theta/2)} = A\int \frac{d\theta}{2\cos(\theta/2)\sin(\theta/2)}$$

Multiplying numerator and denominator by $\cos\theta/2$, the integral become:

$$V = A\int \frac{(1/2)\sec^2(\theta/2)d\theta}{\tan(\theta/2)} = A\ln\left(\tan(\theta/2)\right) + B$$

The bounday conditions are:

(i) $\theta = \theta_1, V = 0$

(ii) $\theta = \theta_2, V = 50V$

From the first boundary conditions:

$$\begin{aligned} V(\theta = \theta_1) = 0 \Longrightarrow 0 &= A\ln\left(\tan(\theta_1/2)\right) + B \Longrightarrow B = -A\ln\left(\tan(\theta_1/2)\right) \\ V(\theta) &= A\ln\left(\tan(\theta/2)\right) - A\ln\left(\tan(\theta_1/2)\right) \\ &= A\ln\left[\frac{(\tan(\theta/2))}{(\tan(\theta_1/2))}\right] = A\ln\left[\frac{(\tan(\theta/2))}{(\tan(\pi/20))}\right] \end{aligned}$$

From the second boundary condition:

$$V(\theta = \theta_2) = 50V \Longrightarrow 50 = A\ln\left[\frac{(\tan(\theta_2/2))}{(\tan(\pi/20))}\right] \Longrightarrow A = 50/\ln\left[\frac{(\tan(\pi/12))}{(\tan(\pi/20))}\right]$$

$$A \approx \cancel{50/0.5 = 100}\ 95.1$$

The potential is then,

$$V(\theta) = \cancel{100}\ 95.1 \ln\left[\frac{(\tan(\theta/2))}{(\tan(\pi/20))}\right] = \cancel{100}\ 95.1 \ln\left[\frac{(\tan(\theta/2))}{0.158}\right]$$

The electric field is calculated as:

$$E = -\frac{1}{r}\frac{dV}{d\theta}\hat{\theta} = -\frac{\cancel{100}\ 95.1}{r}\frac{1}{2}\frac{\sec^2(\theta/2)}{\tan(\theta/2)}\hat{\theta} = -\frac{\cancel{50}\ 47.55}{r}\frac{\sec^2(\theta/2)}{\tan(\theta/2)}\hat{\theta}$$

3. Suppose that the surface of a sphere of radius R carries a potential $V_0(\theta)$. Find the potential inside and out side the sphere and hence the expression for surface charge density.

Solution:
The general solution of Laplace equation for the metallic sphere is:

$$V(r,\theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos\theta)$$

Inside: $r \leq R \Longrightarrow B_l = 0$,otherwise it will blow up

$$\therefore V_1(r,\theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)$$

At $r = R$, $V = V_0$

$$V_0 = \sum_{l=0}^{\infty} A_l R^l P_l(\cos\theta)$$

Using Fourier trick multiply both sides by $P_{l'}(\cos\theta)$ and integrating from 0 to π
.

$$\int_0^{\pi} V_0(\theta) P_{l'}(\cos\theta)\sin\theta d\theta = \sum_{l=0}^{\infty} A_l R^l \int_0^{\pi} P_l(\cos\theta) P_{l'}(\cos\theta)\sin\theta d\theta$$

Where,

$$\int_0^{\pi} P_l(\cos\theta) P_{l'}(\cos\theta)\sin\theta d\theta = \begin{cases} 0, & \text{if } l \neq l' \\ \frac{2}{2l+1}, & \text{if } l = l' \end{cases}$$

Therefore,

$$A_l = \frac{2l+1}{2R^l} \int_0^{\pi} V_0(\theta) P_l(\cos\theta)\sin\theta d\theta$$

The potential inside the sphere is then,

$$V_1(r,\theta) = \sum_{l=0}^{\infty} \frac{2l+1}{2R^l} \left(\int_0^{\pi} V_0(\theta) P_l(\cos\theta)\sin\theta d\theta \right) r^l P_l(\cos\theta)$$

Outside: $r \to \infty \Longrightarrow A_l \to 0$, since they blow up at origin.

$$V_2(r,\theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos\theta)$$

Similarly, At $r = R$, $V = V_0$

$$B_l = \frac{2l+1}{2} R^{l+1} \int_0^{\pi} V_0(\theta) P_l(\cos\theta)\sin\theta d\theta$$

At $r = R$, V is continuous,

$$\sum_{l=0}^{\infty} A_l R^l P_l(\cos\theta) = \sum_{l=0}^{\infty} \frac{B_l}{R^{l+1}} P_l(\cos\theta)$$

We can deduce that,

$$B_l = A_l R^{2l+1}$$

The surface charge density is given as:

$$\left(\frac{\partial V_2}{\partial r} - \frac{\partial V_1}{\partial r}\right)\bigg|_{r=R} = -\frac{\sigma}{\epsilon_0}$$

$$-\sum_{l=0}^{\infty}(l+1)\frac{B_l}{R^{l+2}}P_l\left(\cos\theta\right) - \sum_{l=0}^{\infty} lA_l R^{l-1}P_l\left(\cos\theta\right) = -\frac{\sigma}{\epsilon_0}$$

$$-\sum_{l=0}^{\infty}(l+1)\frac{A_l R^{2l+1}}{R^{l+2}}P_l\left(\cos\theta\right) - \sum_{l=0}^{\infty} lA_l R^{l-1}P_l\left(\cos\theta\right) = -\frac{\sigma}{\epsilon_0}$$

$$\sum_{l=0}^{\infty}(2l+1)A_l R^{l-1}P_l\left(\cos\theta\right) = \frac{\sigma}{\epsilon_0}$$

$$\sigma(\theta) = \epsilon_0 \sum_{l=0}^{\infty}(2l+1)A_l R^{l-1}P_l\left(\cos\theta\right)$$

4. In the above problem, Problem 2, if the potential at the surface $V(\theta) = K\sin^2\theta/2$, ($K$ being a constant) than calculate the potential inside as well as outside the sphere.

Solution:
The potential inside the sphere is:

$$V_1(r,\theta) = \sum_{l=0}^{\infty} A_l r^l P_l\left(\cos\theta\right)$$

At the surface,

$$V(\theta) = K\sin^2\theta/2 = \frac{K}{2}\left(1-\cos\theta\right) = \frac{K}{2}\left[P_0\left(\cos\theta\right) - P_1\left(\cos\theta\right)\right]$$

Therefore,

$(r = R)$ $$\frac{K}{2}\left[P_0\left(\cos\theta\right) - P_1\left(\cos\theta\right)\right] = A_0 P_0\left(\cos\theta\right) + A_1 R P_1\left(\cos\theta\right)$$

Comparing coefficient,

$$A_0 = k/2, A_1 = -k/2R$$

The potential inside the sphere is then,

$$V_1(r,\theta) = \left(\frac{K}{2} - \frac{K}{2R}r\right)\cos\theta$$

The potetntial outside the sphere is :

$$V_2(r,\theta) = \sum_{l=0}^{\infty}\frac{B_l}{r^{l+1}}P_l\left(\cos\theta\right)$$

Then,

$$\frac{K}{2}\left[P_0\left(\cos\theta\right)-P_1\left(\cos\theta\right)\right]=\frac{B_0}{R}P_0\left(\cos\theta\right)+\frac{B_1}{R^2}P_1\left(\cos\theta\right)$$

Comparing the coefficient,

$$B_0=\frac{KR}{2}, B_1=-\frac{KR^2}{2}$$

The potential outside the sphere is then,

$$V_2(r,\theta)=\frac{KR}{2r}-\frac{KR^2}{2r^2}\cos\theta$$

5. A spherical shell of radius R carries uniform surface charge density σ_0 on the upper hemisphere and that of $-\sigma_0$ on the bottom hemisphere. Find the potential inside and outside the sphere up to a minimum of three non zero terms in both the cases.

Solution:

$$V(r,\theta)=\begin{cases}\sum_{l=0}^{\infty}A_l r^l P_l(\cos\theta), & (r\le R),\\ \sum_{l=0}^{\infty}\frac{B_l}{r^{l+1}}P_l(\cos\theta), & (r\ge R),\end{cases}$$

where $B_l=A_lR^{2l+1}$ and

$$A_l=\frac{1}{2\epsilon_0R^{l-1}}\int_0^{\pi}\sigma_0(\theta)P_l(\cos\theta)\sin\theta\,d\theta.$$

Using $x=\cos\theta$, this simplifies to:

$$A_l=\frac{\sigma_0}{2\epsilon_0R^{l-1}}\left[\int_0^1P_l(x)\,dx-\int_{-1}^0P_l(x)\,dx\right].$$

Since $P_l(-x)=(-1)^lP_l(x)$, it follows that:

$$\int_{-1}^0P_l(x)\,dx=(-1)^l\int_0^1P_l(x)\,dx.$$

Thus,

$$A_l=\frac{\sigma_0}{2\epsilon_0R^{l-1}}\left[1-(-1)^l\right]\int_0^1P_l(x)\,dx.$$

For even l, this expression vanishes, so we only need A_1, A_3, A_5.
Evaluating the integrals:

$$\int_0^1P_1(x)dx=\frac{1}{2},\quad\int_0^1P_3(x)dx=-\frac{1}{8},\quad\int_0^1P_5(x)dx=\frac{1}{16}.$$

Therefore,

$$A_1=\frac{\sigma_0}{\epsilon_0}\left(\frac{1}{2}\right),\quad A_3=\frac{\sigma_0}{\epsilon_0R^2}\left(-\frac{1}{8}\right),\quad A_5=\frac{\sigma_0}{\epsilon_0R^4}\left(\frac{1}{16}\right),\quad\text{etc.}$$

Similarly,

$$B_1 = \frac{\sigma_0}{\epsilon_0} R^3 \left(\frac{1}{2}\right), \quad B_3 = \frac{\sigma_0}{\epsilon_0} R^5 \left(-\frac{1}{8}\right), \quad B_5 = \frac{\sigma_0}{\epsilon_0} R^7 \left(\frac{1}{16}\right), \quad \text{etc.}$$

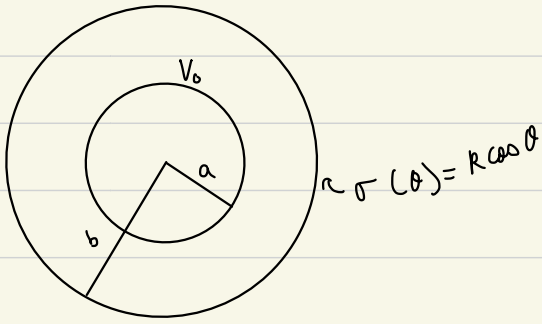
Thus, the potential is given by:

$$V(r,\theta) = \begin{cases} \frac{\sigma_0 r}{2\epsilon_0} \left[P_1(\cos\theta) - \frac{1}{4}\left(\frac{r}{R}\right)^2 P_3(\cos\theta) + \frac{1}{8}\left(\frac{r}{R}\right)^4 P_5(\cos\theta) + \dots \right], & (r \leq R), \\ \frac{\sigma_0 R^3}{2\epsilon_0 r^2} \left[P_1(\cos\theta) - \frac{1}{4}\left(\frac{R}{r}\right)^2 P_3(\cos\theta) + \frac{1}{8}\left(\frac{R}{r}\right)^4 P_5(\cos\theta) + \dots \right], & (r \geq R). \end{cases}$$

6. A conducting sphere of radius a, maintained at a potential V_o, is surrounded by a thin concentric spherical shell of radius b, carrying a surface charge density $\sigma(\theta) = k\cos\theta$, where k is a constant.
(a) Find the potential in region: (i) $r > b$, and (ii) $a < r < b$.
(b) Find the induced surface charge density, $\sigma_{ind}(\theta)$, on the conductor.
(c) What is the total charge of this system? Check that your answer is consistent with the behavior of V at large r.

$(a < r < b)$

$$V_2(r,\theta) = \sum_{k=0}^{\infty} \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}\right) P_\ell(\cos\theta)$$



$$V_3(r,\theta) = \sum_{k=0}^{\infty} \frac{B'_\ell}{r^{\ell+1}} P_\ell(\cos\theta)$$

$(r > b)$

$$V_2(b,\theta) = V_3(b,\theta)$$

$$V_2(a,\theta) = 0$$

$$\left(\frac{\partial V_3}{\partial r} - \frac{\partial V_2}{\partial r}\right)_{r=b} = \frac{-k\cos\theta}{\varepsilon_0} = \frac{-k}{\varepsilon_0} P_\ell(\cos\theta)$$

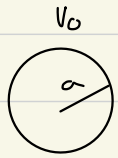
$$A_\ell = \frac{k}{3\varepsilon_0}$$

$$B_1 = \frac{-ka^3}{3\varepsilon_0}$$

$$B'_1 = \frac{-ka^3}{3\varepsilon_0} + \frac{kb^3}{3\varepsilon_0}$$

For under spherical shell

$$V_2(r,\theta) = \frac{k}{3\varepsilon_0} \frac{\cos\theta}{r^2}(r^3 - a^3) + \frac{aV_0}{r}$$



$$V_{out} = \frac{aV_0}{r}$$

$$\sum_{\ell=0}^{\infty} \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos\theta) = \underbrace{V_0 P_0(\cos\theta)}_{V_{out}}$$

$$B_0 = aV_0$$

$$V_3(r,\theta) = \frac{K}{3\varepsilon_0}\left(b^3 - a^3\right)\frac{\cos\theta}{r^2} + \frac{aV_0}{r}$$

$$V(r \gg b) \sim \frac{aV_0}{r}$$